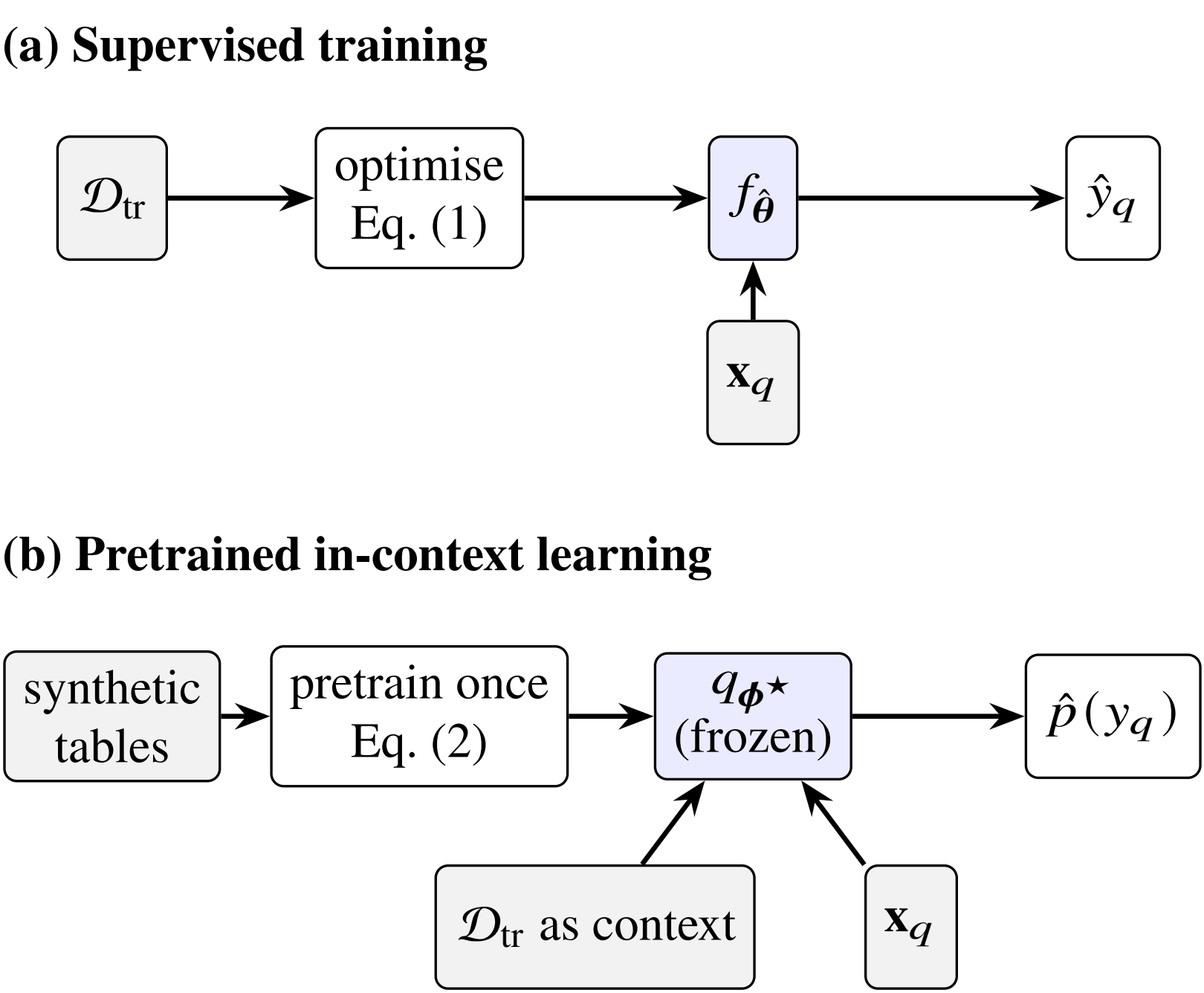

(a) Supervised training
$\mathcal{D}_{\text{tr}}$
optimise Eq. (1)
$f_{\hat{\boldsymbol{\theta}}}$
$\hat{y}_q$
$\mathbf{x}_q$
(b) Pretrained in-context learning
synthetic tables
pretrain once Eq. (2)
$q_{\boldsymbol{\phi}^\star}$ (frozen)
$\hat{p}(y_q)$
$\mathcal{D}_{\text{tr}}$ as context
$\mathbf{x}_q$